ModelIt!

Student Activity Pack

Why Hurricanes Are Getting Angrier

The Science Behind Supercharged Storms

Grade 8 | NGSS MS-ESS2-5, MS-ESS2-6 | Energy and Matter

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What causes hurricanes to form?

2 Why do you think some hurricanes are stronger than others?

3 What do you think happens to hurricane strength if the ocean gets warmer?

4 Draw a diagram showing how you think a hurricane gets its energy.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Sea Surface Temperature | Atmospheric Moisture | Hurricane Wind Speed | Storm Surge Height

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Category 1 Storm

Normal warm ocean temperature, moderate moisture

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Supercharged Storm

Lock Sea Surface Temperature to 90°F

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Cool Water Encounter

Drop temperature from 90°F to 75°F

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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> Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Wind Shear — Strong winds at different altitudes that can tear apart hurricanes

Rainfall Intensity — Amount of rain the storm produces, linked to moisture

Coastal Population Density — Number of people in the storm's path, affecting damage

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why are hurricanes getting stronger and more destructive? Use evidence from your simulation to support your answers.

1 Why is a Category 5 hurricane over warm water more dangerous than one over cool water?

2 How does the concept of 'rapid intensification' make hurricane forecasting harder?

3 If ocean temperatures continue to rise, what does your model predict for future storms?

4 Why do hurricanes weaken when they hit land?

5 Should people continue to build communities in hurricane-prone coastal areas? Why or why not?

8 STEM Challenge: Design a Hurricane-Resilient Community

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YOUR ENGINEERING MISSION

Design a coastal community plan that can withstand Category 5 hurricanes based on what your model reveals about storm behavior.

THE SCENARIO

A coastal city that experienced a devastating hurricane has hired your team to redesign their community to survive future storms that may be even stronger.

Engineering Design Thinking

1. WHERE SHOULD BUILDINGS BE LOCATED RELATIVE TO THE COAST?
2. WHAT BUILDING MATERIALS AND DESIGNS RESIST HURRICANE FORCES?
3. HOW WILL YOU PROTECT AGAINST STORM SURGE FLOODING?
4. WHAT EMERGENCY SYSTEMS DOES YOUR COMMUNITY NEED?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Meteorologists and Climate Scientists study weather patterns and develop forecasting models. Hurricane Hunters fly INTO storms to collect data! They earn $70,000-$120,000/year.